Es 1. C.E.

a) $\frac{1}{x^2}$ C.E. $x^2 \neq 0 \Rightarrow x \neq 0$

b) $\frac{1}{3x+12}$ C.E. $3x+12 \neq 0$

$3x \neq -12$

$x \neq -\frac{12}{3}$

$x \neq -4$

c) $\frac{1}{4+x^2}$ C.E. $\forall x \in \mathbb{R}$

d) $\frac{x-1}{16-x^2} = \frac{x-1}{(4-x)(4+x)}$

C.E. $4-x \neq 0 \Rightarrow -x \neq -4 \Rightarrow x \neq +4$

$4+x \neq 0 \Rightarrow x \neq -4$

Es 2. Semplificazioni

$\frac{2x^2-x}{x^2-2x^3} = \frac{x(2x-1)}{x^2(1-2x)} = \frac{x(2x-1)}{-x^2(2x-1)}$

C.E. • $-x^2 \neq 0 \Rightarrow x \neq 0$

• $2x-1 \neq 0 \Rightarrow x \neq \frac{1}{2}$

$\frac{x(2x-1)}{-x^2(2x-1)} = -\frac{1}{x}$

b) $\frac{x-3+2x}{x^2-1} = \frac{3x-3}{(x+1)(x-1)} = \frac{3(x-1)}{(x+1)(x-1)}$

C.E. $x+1 \neq 0 \Rightarrow x \neq -1$

$x-1 \neq 0 \Rightarrow x \neq +1$

$\frac{3(x-1)}{(x+1)(x-1)} = \frac{3}{x+1}$

c) $\frac{x^4+2x^2}{x^4+4x^2+4} = \frac{x^2(x^2+2)}{(x^2+2)^2}$

C.E. $(x^2+2)^2 \neq 0 \quad \forall x \in \mathbb{R}$

↳ è sempre $\neq 0$

$\frac{x^2(x^2+2)}{(x^2+2)^2} = \frac{x^2}{x^2+2}$

Es 3

a) $\frac{x^2+3x}{x^2-x^4} \cdot \frac{x^2+x^3}{x^3+6x^2+9x} =$

$= \frac{x(x+3)}{x^2(1-x^2)} \cdot \frac{x^2(1+x)}{x(x^2+6x+9)}$

$= \frac{x(x+3)}{-x^2(x+1)(x-1)} \cdot \frac{x^2(x+1)}{x(x+3)^2} =$

C.E. • $x \neq 0$

• $x \neq -3$

• $x \neq 1$

• $x \neq -1$

$= \frac{x(x+3)}{-x^2(x+1)(x-1)} \cdot \frac{x^2(x+1)}{x(x+3)^2} =$

$= -\frac{1}{(x-1)(x+3)}$

b) $\frac{x^4+x^2}{x^5-x^3} : \frac{x^3+x}{x^3+x^2}$

$= \frac{x^2(x^2+1)}{x^3(x+1)(x-1)} : \frac{x(x^2+1)}{x^2(x+1)}$

= C.E. • $x \neq 0$

• $x \neq -1$

• $x \neq +1$

• $\forall x \in \mathbb{R}$

$\frac{x^2(x^2+1)}{x^3(x+1)(x-1)} \cdot \frac{x^2(x+1)}{x(x^2+1)} = \frac{1}{x-1}$